1 Z-Test:

One-Sample Z-Test for a Mean	**Two-Sample Z-Test for Means**
Problem it solves: Tests whether a population mean differs from a specified reference value.	**Problem it solves:** Tests whether the means of two independent populations differ.
Null Hypothesis (H_0): $H_0 : \mu = \mu_0$	**Null Hypothesis (H_0):** $H_0 : \mu_1 - \mu_2 = 0$
Alternative Hypothesis (H_1): $H_1 : \mu \neq \mu_0$	**Alternative Hypothesis (H_1):** $H_1 : \mu_1 - \mu_2 \neq 0$
Test Statistic: $Z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$	**Test Statistic:** $Z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$
Symbol Meanings: $\bar{x}$: Sample mean μ_0: Mean specified under H_0 σ: Known population standard deviation n: Sample size	**Symbol Meanings:** $\bar{x}_1, \bar{x}_2$: Sample means σ_1, σ_2: Known population standard deviations n_1, n_2: Sample sizes
Example: Is the average delivery time different from 30 minutes?	**Example:** Do two delivery systems have different average delivery times?

One-Proportion Z-Test	**Two-Proportion Z-Test**
Problem it solves: Tests whether a population proportion differs from a specified reference proportion.	**Problem it solves:** Tests whether the proportions of two independent populations differ.
Null Hypothesis (H_0): $$H_0 : p = p_0$$	**Null Hypothesis (H_0):** $$H_0 : p_1 - p_2 = 0$$
Alternative Hypothesis (H_1): $$H_1 : p \neq p_0$$	**Alternative Hypothesis (H_1):** $$H_1 : p_1 - p_2 \neq 0$$
Test Statistic: $$Z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$$	**Test Statistic:** $$Z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$
Symbol Meanings: $\hat{p}$: Sample proportion p_0: Proportion specified under H_0 n: Sample size	**Symbol Meanings:** $\hat{p}_1, \hat{p}_2$: Sample proportions n_1, n_2: Sample sizes $\hat{p}$: Pooled sample proportion under H_0 $$\hat{p} = \frac{x_1 + x_2}{n_1 + n_2}$$ x_1, x_2: Number of successes in each sample
Example: Is the population click-through rate different from 50%?	**Example:** Do two website versions have different conversion rates?

2 T-Test:

One-Sample T-Test	**Independent Two-Sample T-Test**	**Paired-Sample T-Test**
Problem it solves: Tests whether a population mean differs from a specified reference value.	**Problem it solves:** Tests whether the means of two independent populations differ.	**Problem it solves:** Tests whether the mean difference between paired observations differs from zero or another specified value.
Null Hypothesis: $H_0 : \mu = \mu_0$	**Null Hypothesis:** $H_0 : \mu_1 - \mu_2 = 0$	**Null Hypothesis:** $H_0 : \mu_d = 0$
Alternative Hypothesis: $H_1 : \mu \neq \mu_0$	**Alternative Hypothesis:** $H_1 : \mu_1 - \mu_2 \neq 0$	**Alternative Hypothesis:** $H_1 : \mu_d \neq 0$
Test Statistic: $t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$	**Test Statistic (Welch's):** $t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$	**Test Statistic:** $t = \frac{\bar{d} - \mu_{d,0}}{s_d/\sqrt{n}}$
Degrees of Freedom: $df = n - 1$	**Degrees of Freedom:** Welch's approximation: $df \approx \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{(s_1^2/n_1)^2}{n_1 - 1} + \frac{(s_2^2/n_2)^2}{n_2 - 1}}$	**Degrees of Freedom:** $df = n - 1$
Symbol Meanings: $\bar{x}$: Sample mean μ_0: Hypothesized population mean s: Sample standard deviation n: Sample size	**Symbol Meanings:** $\bar{x}_1, \bar{x}_2$: Sample means s_1, s_2: Sample standard deviations n_1, n_2: Sample sizes	**Symbol Meanings:** $\bar{d}$: Mean of paired differences s_d: Standard deviation of paired differences n: Number of pairs $\mu_{d,0}$: Hypothesized mean difference
Example: Is the average delivery time different from 30 minutes?	**Example:** Do two independent delivery systems have different average delivery times?	**Example:** Do customers spend different amounts before and after a loyalty program?
When to use: One sample compared with a reference mean; population standard deviation is unknown.	**When to use:** Two independent groups; population standard deviations are unknown. Welch's test does not assume equal population variances.	**When to use:** The same individuals measured twice, or observations matched into pairs. Analyze the difference within each pair.

3 Chi-Square Test:

Chi-Square Test of Independence	**Chi-Square Goodness-of-Fit Test**
Problem it solves: Tests whether two categorical variables are associated with each other.	**Problem it solves:** Tests whether observed category frequencies differ significantly from the frequencies expected under a specified distribution.
Null Hypothesis (H_0): The two categorical variables are independent (there is no association between them).	**Null Hypothesis (H_0):** The population follows the specified distribution of category proportions.
Alternative Hypothesis (H_1): The two categorical variables are associated.	**Alternative Hypothesis (H_1):** The population does not follow the specified distribution.
Test Statistic: $$\chi^2 = \sum_{i=1}^{r}\sum_{j=1}^{c}\frac{(O_{ij}-E_{ij})^2}{E_{ij}}$$	**Test Statistic:** $$\chi^2 = \sum_{i=1}^{k}\frac{(O_i-E_i)^2}{E_i}$$
Expected Frequency: $$E_{ij} = \frac{(\text{Row Total}_i)(\text{Column Total}_j)}{\text{Grand Total}}$$	**Expected Frequency:** $$E_i = np_i$$
Degrees of Freedom: $$df = (r-1)(c-1)$$	**Degrees of Freedom:** $$df = k-1-m$$
Symbol Meanings: O_{ij}: Observed count in row i, column j E_{ij}: Expected count under independence r: Number of rows c: Number of columns	**Symbol Meanings:** O_i: Observed count in category i E_i: Expected count in category i n: Total sample size p_i: Hypothesized proportion for category i k: Number of categories m: Number of distribution parameters estimated from the data
Example: Is customer churn associated with subscription type?	**Example:** Do customer preferences across four product categories follow specified proportions?

4 ANOVA:

One-Way ANOVA	Two-Way ANOVA	Repeated-Measures ANOVA
Problem it solves: Tests whether the means of three or more independent groups differ, using one categorical factor.	**Problem it solves:** Tests how two categorical factors relate to a numerical outcome, including their individual effects and interaction.	**Problem it solves:** Tests whether the means of three or more related conditions or repeated measurements differ.
Null Hypothesis (H_0): $H_0 : \mu_1 = \mu_2 = \cdots = \mu_k$	**Null Hypotheses (H_0):** For factor A: H_0 : No main effect of A For factor B: H_0 : No main effect of B For interaction: H_0 : No A $\times$ B interaction	**Null Hypothesis (H_0):** $H_0 : \mu_1 = \mu_2 = \cdots = \mu_k$ The population means are equal across repeated conditions.
Alternative Hypothesis (H_1): H_1 : At least one mean differs	**Alternative Hypotheses (H_1):** At least one factor has a main effect, or the effect of one factor depends on the level of the other factor.	**Alternative Hypothesis (H_1):** At least one repeated-condition population mean differs.
Test Statistic: $F = \frac{MS_{\text{between}}}{MS_{\text{within}}}$	**Test Statistic:** Separate F-statistics test each effect: $F_A = \frac{MS_A}{MS_E}$ $F_B = \frac{MS_B}{MS_E}$ $F_{AB} = \frac{MS_{AB}}{MS_E}$	**Test Statistic:** $F = \frac{MS_{\text{condition}}}{MS_{\text{error}}}$ The error term accounts for variation associated with subjects and conditions under the repeated-measures model.
Symbol Meanings: k: Number of groups MS_{between}: Mean square between groups MS_{within}: Mean square within groups	**Symbol Meanings:** MS_A: Mean square for factor A MS_B: Mean square for factor B MS_{AB}: Mean square for the interaction MS_E: Appropriate residual mean square	**Symbol Meanings:** $MS_{\text{condition}}$: Mean square for conditions MS_{error}: Mean square for the appropriate repeated-measures error term k: Number of conditions
Example: Do three marketing campaigns produce different average sales?	**Example:** Do marketing campaign and season affect sales, and does campaign effectiveness depend on season?	**Example:** Do the same customers spend different amounts before, during, and after a loyalty program?
When to use: One numerical outcome, one categorical factor, and independent observations across groups.	**When to use:** One numerical outcome and two categorical factors. Observations must satisfy the design-specific independence and model assumptions.	**When to use:** The same subjects are measured under three or more conditions or at multiple time points. Repeated observations within a subject are related.

5 Pearson Correlation Test:

Component	**Pearson Correlation Test**
Purpose	Tests whether two variables have a significant linear relationship.
Null Hypothesis (H_0)	$\rho = 0$ (no linear correlation in the population).
Alternative Hypothesis (H_1)	$\rho \neq 0$ (a linear correlation exists).
Correlation Coefficient	$r = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum(x_i - \bar{x})^2 \sum(y_i - \bar{y})^2}}$
Test Statistic	$t = r\sqrt{\frac{n-2}{1-r^2}}$
Degrees of Freedom	$df = n - 2$
Interpretation	r ranges from -1 to $+1$. Positive values indicate a positive linear relationship, while negative values indicate a negative linear relationship. Values near zero indicate a weak or absent linear relationship.